POSITIVE COMPACT PROBES AND FREE RESPONSE COEFFICIENTS FOR LATTICE QED IN THREE DIMENSIONS

Abstract. We construct compactly supported real two-form probes with positive fourth response for the massive doubled Dirac operator in three dimensions. The proof gives the local large-mass coefficient $1/(240\pi M^5)$ and a remainder of order M^{-7}. For the chamber lattice regulator, the fourth logarithmic determinant derivative converges to this continuum trace with error $O(\eta^{1/4})$ at fixed physical mass, uniformly in the torus size; the remaining volume error decays exponentially. A cancellation between the two-vertex and contact terms gives uniform bounds on the second and fourth ordinary determinant derivatives. The constants are uniform under real translations of the compact probe.

Contents

1. Introduction 1
2. The local continuum response 3
3. The chamber symbol and determinant derivatives 8
4. Comparison of the fourth traces 10
5. Symmetry and the second-order response 13
References 18

1. Introduction

Let σ_ν be the Pauli matrices and $\gamma_\nu = \operatorname{diag}(\sigma_\nu, -\sigma_\nu)$, $1 \le \nu \le 3$. Thus $\gamma_\nu\gamma_\mu + \gamma_\mu\gamma_\nu = 2\delta_{\mu\nu}$. On $L^2(\mathbb{R}^3;\mathbb{C}^4)$, for $M > 0$ and real $a \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$, write

$$S_M = (M + \gamma\cdot\partial)^{-1}, \qquad V_a = i\gamma\cdot a, \qquad q_{4,M}(a) = -\tfrac{1}{4}\operatorname{tr}[(S_M V_a)^4].$$

For a real compact smooth two-form β, use the Euclidean codifferential $(d^*\beta)_\nu = -\sum_{\mu\neq\nu}\partial_\mu\beta_{\mu\nu}$, put $f = dd^*\beta$, and define, at fixed mass $\bar m > 0$,

$$b(f) = -6\operatorname{tr}[(S_{\bar m} i\gamma\cdot d^*\beta)^4] = 24 q_{4,\bar m}(d^*\beta). \tag{1}$$

This depends only on f: two choices of $d^*\beta$ with the same exterior derivative differ by a compactly supported smooth one-form that is closed and coclosed, hence harmonic and therefore zero.

Theorem A (A positive continuum probe)**.** For every real $a \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$ and $M > 0$, the operator $(S_M V_a)^4$ is trace class and its trace is real. There is $C_a < \infty$ such that, for $M \ge 1$,

$$\left| q_{4,M}(a) - \frac{1}{240\pi M^5}\int_{\mathbb{R}^3} |\operatorname{curl} a|^4 \,\mathrm{d}x \right| \le C_a M^{-7}. \tag{2}$$

In particular, for every $\bar m > 0$ there is a real $\beta \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ for which $f = dd^*\beta$ satisfies $b(f) > 0$. ■□

The sign in (1) is obtained by dilating the compact probe after evaluating the large-mass coefficient. The leading local term is the three-dimensional analogue of the quartic field-strength term in the Heisenberg–Euler effective Lagrangian [HE36]. The coefficient and remainder in theorem A are proved here by the continuum argument of [26c, §17].

1.1. **The lattice response.** Fix $0 < \rho < 1$ and $0 < \bar{m} \le 1$. Let β be a real nonzero compactly supported smooth two-form, and put $a = d^*\beta$, $f = dd^*\beta$. Thus $a_\nu = -\sum_{\mu\neq\nu} \partial_\mu \beta_{\mu\nu}$. Let Q_β be a closed coordinate cube of least side D_β containing $\operatorname{supp}\beta$; this side includes no buffer. For $0 < \eta \le 1$ define

$$J_\eta v(y) = \eta^{-3} \int_{[-\eta/2,\eta/2]^3} v(y+z)\,\mathrm{d}z, \qquad \delta_{\eta,\mu} v(y) = \frac{v(y+\eta e_\mu/2) - v(y-\eta e_\mu/2)}{\eta}.$$

The interpolating functions

$$a^0_{\eta,\nu}(y) = -\sum_{\mu\neq\nu} \delta_{\eta,\mu} J_\eta \beta_{\mu\nu}(y) \tag{3}$$

are defined without periodization. If $n = R/\eta \in \mathbb{N}$, $n \ge 2$, the lattice one-cochain on the physical torus is

$$a_{\eta,\nu}(\eta c) = \sum_{z\in\mathbb{Z}^3} a^0_{\eta,\nu}(\eta c + \eta e_\nu/2 + Rz). \tag{4}$$

It is the adjoint lattice coboundary of the cube-averaged probe at plaquette centers. In lattice units the external connection is $B_t(c, c+e_\nu) = t\eta a_{\eta,\nu}(\eta c)$, extended with the orientation sign to reversed bonds. The free mass in these units is $m = \bar{m}\eta$.

The chamber labels are the 48 signed permutation matrices $g \in O_{\mathrm{cub}} = O(3) \cap \mathrm{GL}(3,\mathbb{Z})$. Let s_0 change the sign of the third coordinate, let s_1, s_2 interchange coordinates $(2,3)$ and $(1,2)$, respectively, and let r_{e_1} change the sign of the first coordinate. Set

$$n_0 = e_3, \quad n_1 = (e_2 - e_3)/\sqrt{2}, \quad n_2 = (e_1 - e_2)/\sqrt{2}, \quad n_3 = -e_1,$$
$$(w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6).$$

For a spinor $\psi : \mathbb{Z}^3 \times O_{\mathrm{cub}} \to \mathbb{C}^4$ with $\psi(c + ne_\nu, g) = -\psi(c,g)$, put $M_i(g) = \frac{w_i}{2}(\rho + \gamma \cdot g n_i)$ and define

$$\begin{aligned}(D^{\mathrm{ch}}[B]\psi)(c,g) = \sum_{i=0}^{2} & M_i(g)\big(\psi(c,g) - \psi(c, gs_i)\big) \\ & + M_3(g)\big(\psi(c,g) - e^{iB(c,c+ge_1)}\psi(c+ge_1, gr_{e_1})\big).\end{aligned} \tag{5}$$

Here B is an $n\mathbb{Z}^3$-periodic cochain on the oriented nearest-neighbor edges of the covering lattice, with $B(c',c) = -B(c,c')$ for $c, c' \in \mathbb{Z}^3$. The operator is determined by its values for $c \in \{0,\dots,n-1\}^3$; when $n = 2$, edges with the same quotient endpoints remain distinct. This is the chamber operator of [26b, §2] in lattice units. The spinor pairing uses counting measure in c and $48^{-1}\sum_g$ in the labels.

Let

$$N_0(t) = \frac{\det(D^{\mathrm{ch}}[B_t] + m)}{\det(D^{\mathrm{ch}}[0] + m)}, \qquad c_j = N_0^{(j)}(0), \qquad a_{\mathrm{ch}} = (\log N_0)''(0), \quad b_{\mathrm{ch}} = (\log N_0)^{(4)}(0). \tag{6}$$

The logarithm is the holomorphic branch equal to zero at the origin. Its denominator is nonzero: the real part of $D^{\mathrm{ch}}[0] + m$ is at least $m > 0$. For comparison with the charge convention, for $e > 0$ put $q = e\eta^{1/2}$; the normalized probe is $\widehat{a} = \eta^{1/2} a_\eta$, so $B_t = q(t/e)\widehat{a}$. The free ratio and its coefficients are independent of e.

On $T_R = \mathbb{R}^3/R\mathbb{Z}^3$ use spinors antiperiodic in all three directions. Their momentum set is $\mathcal{P}_R = \frac{2\pi}{R}(\mathbb{Z} + \frac{1}{2})^3$, and S^R is the multiplier $s_{\bar{m}}(P)$ there, where $s_m(p) = (m + i\gamma\cdot p)^{-1}$. For $R \ge D_\beta + 1$ let $a^R = \sum_z a(\,\cdot + Rz)$, and put

$$b_R(f) = -6\,\mathrm{Tr}(S^R i\gamma \cdot a^R)^4, \qquad b(f) = -6\,\mathrm{Tr}(S_{\bar{m}} i\gamma\cdot a)^4. \tag{7}$$

Theorem B (Free lattice response)**.** For the operator and sampled probe above, fix $0 < \rho < 1$, $0 < \bar{m} \le 1$, and real nonzero $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$. For all $0 < \eta \le 1$ and $n = R/\eta \in \mathbb{N}$ with $n \ge 2$, the following assertions hold.

If $R \geq D_\beta + 1$, the operators $S^R i\gamma \cdot a^R$ and $S_{\bar m} i\gamma \cdot a$ belong to the Schatten class of order four; their fourth-power traces are real. There are constants $C_1(\rho, \bar m, \beta)$ and $C_2(\bar m, \beta)$ such that

$$|b_{\mathrm{ch}}(\eta, R) - b_R(f)| \leq C_1 \eta^{1/4},$$
$$|b_R(f) - b(f)| \leq C_2 e^{-\bar m(R - D_\beta)}, \qquad R \geq D_\beta + 1, \quad 0 < \eta \leq 1, \quad R/\eta \in \mathbb{N}. \tag{8}$$

The constants are unchanged by every real translate of the probe. In particular $b_{\mathrm{ch}}(\eta, R)$ tends to $b(f)$ whenever $\eta \to 0$ and $R \to \infty$.

The function N_0 is entire, even, and positive on the real axis. Its derivatives at zero are real and satisfy

$$c_{2j+1} = 0, \qquad c_2 = a_{\mathrm{ch}}, \qquad c_4 = b_{\mathrm{ch}} + 3c_2^2.$$

There is $C_{\rho,\bar m,\beta} < \infty$, unchanged by real translations of β, such that

$$|c_2| \leq C_{\rho,\bar m,\beta} \quad (R \geq 1), \qquad |c_4| \leq C_{\rho,\bar m,\beta} \quad (R \geq D_\beta + 1).$$

All coefficients are independent of e in the charge convention above. They are unchanged by translating the probe through $\eta\mathbb{Z}^3$ or by replacing its external cochain by a cubic reflected cochain. ■□

1.2. **The proof and its use.** For a probe chosen by theorem A, the errors in (8) are eventually smaller than its positive coefficient. The choice of probe precedes every interacting comparison. The free estimates themselves apply to every real compact smooth probe, regardless of the sign of $b(f)$.

The continuum proof differentiates the fourth trace in the mass. Its diagonal kernel admits a local expansion, and gauge covariance reduces the leading term to the antisymmetric first derivative of the connection. A constant-field heat kernel evaluates that term; integration in the mass and dilation give theorem A. The kernel regularity and coefficientwise inverse identities are proved in section 2.

For the lattice comparison, the chamber inverse splits into a Dirac inverse on constant labels and a uniformly bounded remainder. Repeated differentiation of a bond phase gives contact terms. The contact traces tend to zero, while the four first-order vertices converge in Schatten norm. Fourier aliases and the discarded high momenta are bounded uniformly in the torus size. The comparison with infinite space then follows from the exponential decay of the Yukawa images. These comparisons are proved in section 4.

The second derivative needs a separate cancellation. Its two-vertex trace and contact trace form a polarization kernel whose momentum integral vanishes to second order at zero. Pairing its quadratic bound and the finite-volume quadrature error with the decaying probe transform yields the uniform bound for c_2. The identity $c_4 = b_{\mathrm{ch}} + 3c_2^2$ then gives the bound for the ordinary fourth derivative. The cancellation is proved in section 5.2, and section 5.3 completes the proof of theorem B.

The proofs reproduce the continuum and chamber response arguments of [26c, §17–18]. The chamber construction and its free Schur-complement estimates are imported from that manuscript with precise references below. The present results concern the free determinant at zero background connection.

We use Euclidean norms and the Fourier convention $\widehat v(P) = \int_{\mathbb{R}^3} v(x) e^{-iP\cdot x}\,\mathrm{d}x$. Write $|\beta|_{C^j} = \max_{\mu<\nu,\,|\alpha|\leq j} \|\partial^\alpha \beta_{\mu\nu}\|_\infty$. For compact operators, $\|\cdot\|_p$ denotes the Schatten norm; bounded multiplication, completeness, trace cyclicity and Hölder's inequality are used in the forms of [Pon25, slides 16, 53–56]. Finite-dimensional traces include all spin and chamber labels.

2. The local continuum response

We prove theorem A. The mass derivative has a five-propagator kernel whose diagonal is continuous. A compact gauge transformation removes the constant connection at each point. Quartic homogeneity and parity then leave only the fourth power of the field strength at leading order. The constant-field calculation determines its coefficient.

2.1. **Kernels and differentiation in the mass.** The kernel and Fourier multiplier of S_M are

$$S_M(z) = (M - \gamma\cdot\partial)\frac{e^{-M|z|}}{4\pi|z|}, \tag{9}$$

$$\widehat{S}_M(p) = \frac{M - i\gamma\cdot p}{M^2+|p|^2}, \qquad |\widehat{S}_M(p)| = (M^2+|p|^2)^{-1/2}. \tag{10}$$

In particular,

$$|S_M(z)| \le \frac{e^{-M|z|}(2M|z|+1)}{4\pi|z|^2}, \qquad S_M(z) = M^2 S_1(Mz), \qquad \|S_M\|_{L^1} \le 3/M.$$

The operator $G = S_M V_a S_M$ is Hilbert–Schmidt. Indeed, its momentum kernel, with measure $(2\pi)^{-3}\mathrm{d}p$, is $\widehat{S}_M(p) i\gamma\cdot\widehat{a}(p-q)\widehat{S}_M(q)$. Writing $u(p) = (M^2+|p|^2)^{-1}$, the matrix norm bound and Young's inequality give

$$\|G\|_{\mathrm{HS}}^2 \le 8(2\pi)^{-6}\int u(p)(|\widehat{a}|^2 * u)(p)\,\mathrm{d}p \le 8(2\pi)^{-3}\|u\|_2^2\|a\|_2^2 = \frac{\|a\|_2^2}{\pi M}.$$

Here $\|u\|_2^2 = \pi^2/M$. Since $(S_M V_a)^4 = G V_a G V_a$,

$$|q_{4,M}(a)| \le \frac{\|a\|_2^2\|a\|_\infty^2}{4\pi M}. \tag{11}$$

The matrix $\gamma_5 = \left(\begin{smallmatrix} 0 & 1_2 \\ 1_2 & 0 \end{smallmatrix}\right)$ conjugates S_M to S_M^* and V_a to $V_a^* = -V_a$. Conjugation of the fourth power and cyclicity of the trace therefore give a real trace. Cyclicity here involves a bounded factor and the trace-class product GV_aG, so it does not presume that $S_M V_a$ itself is trace class.

We will evaluate the mass derivative on the diagonal of a kernel with five propagators. This requires continuity at the diagonal, including for the linear connections used below. For either a compact smooth connection or a linear connection, $|a(z)| \le c_a(1+|z|)$ and a is Lipschitz. Choose $s(z) = K|z|^{-2}e^{-M|z|/2}$ so that $|S_M(z)|(1+|z|)^{24} \le s(z)$. Its convolution powers obey

$$s^{*2}(z) \le C|z|^{-1}e^{-M|z|/2}, \quad s^{*2}(z) \le C|z|^{-3/2}, \quad s^{*3}(z) \le C|z|^{-1/2}. \tag{12}$$

For example, the bound for the second convolution follows by factoring out $e^{-M|z|/2}$ and scaling $\int |z-w|^{-2}|w|^{-2}\,\mathrm{d}w = C|z|^{-1}$; the third follows by convolving $|z|^{-2}$ and $|z|^{-3/2}$. Thus $s^{*2} \in L^2$, $s^{*4} = s^{*2} * s^{*2}$ is bounded and continuous, and so is $s^{*5} = s * s^{*4}$.

Let $R_a^{(j)}(x,y)$ be the integral kernel of $(-1)^j(S_M V_a)^j S_M$, initially off the diagonal, for $0 \le j \le 4$. Distributing the factors $1+|z_l|$ along the chain of propagators gives

$$|R_a^{(j)}(x,y)| \le [c_a(1+|x|)]^j s^{*(j+1)}(x-y). \tag{13}$$

These integrals act as the indicated operators on Schwartz spinors; absolute integration against such a spinor justifies every composition. The fourth kernel extends continuously to all (x,y). To see this, use the four independent increments of the first three and last propagators. Changing (x,y) then changes four Lipschitz factors and the middle propagator. On $|x|,|y|,|x'|,|y'| \le r$, the resulting difference is bounded by

$$C_{a,r}\big((|x-x'|+|y-y'|)\|s^{*5}\|_\infty + \|s^{*4}\|_\infty \|S_M - S_M(\,\cdot\,+(x'-y')-(x-y))\|_{L^1}\big). \tag{14}$$

This tends to zero. Define $r_4(x;M,a) = \operatorname{tr} R_a^{(4)}(x,x)$ using this extension.

For compact a, differentiation in M now gives

$$\frac{\mathrm{d}}{\mathrm{d}M} q_{4,M}(a) = \int r_4(x;M,a)\,\mathrm{d}x, \qquad q_{4,M}(a) \longrightarrow 0 \quad (M\longrightarrow\infty). \tag{15}$$

To justify this differentiation, write the trace as the absolutely convergent four-vertex integral

$$-\tfrac{1}{4}\int \operatorname{tr}\big[S_M(z_4-z_1)V_a(z_1)S_M(z_1-z_2)V_a(z_2)S_M(z_2-z_3)V_a(z_3)S_M(z_3-z_4)V_a(z_4)\big]\,\mathrm{d}z_1\cdots\mathrm{d}z_4.$$

Its absolute integral is at most $4\|a\|_\infty^3\|a\|_{L^1}s^{*4}(0)$ before the factor $1/4$. On any compact interval of positive masses, S_M and $\partial_M S_M$ have a common majorant of the same form. Since $\partial_M S_M = -S_M * S_M$, the four differentiated terms have equal integrals by cyclicity. The factor $-1/4$ and the minus sign in this derivative cancel. Inserting the convolution yields the diagonal five-propagator integral in (15); its absolute integral is bounded by $4\|a\|_\infty^3\|a\|_{L^1}s^{*5}(0)$. This also proves continuous differentiability. The limit at infinity follows from (11).

2.2. **The local large-mass expansion.** Write

$$V_j(x;w) = i\gamma\cdot\sum_{|\alpha|=j}\frac{w^\alpha}{\alpha!}\partial^\alpha a(x).$$

The substitution $z_l = x + M^{-1}w_l$ in the diagonal kernel gives

$$\begin{aligned} r_4(x;M,a) = M^{-2}\operatorname{tr}\int\, & S_1(-w_1)V_a(x+M^{-1}w_1)S_1(w_1-w_2)V_a(x+M^{-1}w_2)\\ & \cdot S_1(w_2-w_3)V_a(x+M^{-1}w_3)S_1(w_3-w_4)V_a(x+M^{-1}w_4)S_1(w_4)\,\mathrm{d}w. \end{aligned} \tag{16}$$

Let $\rho_j(x;a)$ be the sum of these integrals with the four connection factors replaced by $V_{j_l}(x;w_l)$ and $j_1+\cdots+j_4=j$. Taylor expansion through total order five gives

$$r_4(x;M,a) = \sum_{j=0}^{5} M^{-2-j}\rho_j(x;a) + O_a(M^{-8}), \qquad M\ge 1, \tag{17}$$

uniformly in x. The remainder contains four factors bounded by $C\|a\|_{C^6}(1+|w_l|)^6$ and at least the factor M^{-6}. With $w_0=0$, $\prod_l(1+|w_l|)^6 \le \prod_l(1+|w_l-w_{l-1}|)^{24}$. Use the weighted majorant at unit mass for the five factors S_1. Its fifth convolution at zero bounds the remainder integral by a constant independent of M. The coefficients vanish off $\operatorname{supp} a$. There is also a spatial tail bound

$$\int_{\operatorname{dist}(x,\operatorname{supp} a)\ge 1} |r_4(x;M,a)|\,\mathrm{d}x \le C_a e^{-M}. \tag{18}$$

Indeed, the first and last propagators each travel at least $d=\operatorname{dist}(x,\operatorname{supp} a)$; their bound is Me^{-Md} for $d\ge 1$. Integrating the three intervening propagators gives

$$|r_4(x;M,a)| \le 108\|a\|_\infty^4|\operatorname{supp} a|M^{-1}e^{-2Md}.$$

For $M,d\ge 1$, $2Md\ge M+d$, which proves (18).

The reduction to field strength must be justified at the level of this diagonal kernel. Let $\lambda\in C_c^\infty(\mathbb{R}^3)$ be real. On Schwartz spinors, as a formal power series in u,

$$(M+\gamma\cdot\partial+ui\gamma\cdot(a+d\lambda))^{-1} = e^{-iu\lambda}(M+\gamma\cdot\partial+ui\gamma\cdot a)^{-1}e^{iu\lambda}.$$

All coefficients preserve Schwartz space. Comparing the coefficient of u^4 gives, almost everywhere off the diagonal,

$$R^{(4)}_{a+d\lambda}(x,y) = \sum_{j=0}^{4} R_a^{(j)}(x,y)\frac{[-i(\lambda(x)-\lambda(y))]^{4-j}}{(4-j)!}. \tag{19}$$

For $j=0,1,2,3$, the bounds in (12)–(13) show that these terms are respectively $O(|x-y|^2)$, $O(|x-y|^2)$, $O(|x-y|^{3/2})$, and $O(|x-y|)$ locally in x. Continuity of the fourth kernel therefore implies

$$r_4(x;M,a+d\lambda) = r_4(x;M,a).$$

Uniqueness of the finite asymptotic expansion in (17) gives the same invariance for ρ_j, $0\le j\le 5$.

Fix x. If $p^{(k)}(z)$ is the homogeneous part of degree k of the fifth-order Taylor polynomial of $a(x+z)$, choose a compact λ equal near x to

$$\lambda(x+z) = -\sum_{k=0}^{5} \frac{z \cdot p^{(k)}(z)}{k+1}.$$

Euler's identity gives $z \cdot (a + d\lambda)^{[5]}(x+z) = 0$ for the Taylor polynomial. Thus the transformed constant term vanishes, and its first derivative is the antisymmetric matrix $F_{\nu\mu}(x)/2$, where $F_{\nu\mu} = \partial_\nu a_\mu - \partial_\mu a_\nu$. Each term defining ρ_j has four connection factors. Hence it vanishes for $j < 4$, and at order four only $(j_1, j_2, j_3, j_4) = (1,1,1,1)$ remains. Put

$$A_{F,\mu}(z) = \tfrac{1}{2} \sum_\nu z_\nu F_{\nu\mu}, \qquad \Psi(F) = r_4(0; 1, A_F).$$

Then $\rho_4(x; a) = \Psi(F(x))$. The odd coefficients vanish as well: substitution $w_l \mapsto -w_l$ and conjugation by γ_5 leave the trace unchanged and multiply it by $(-1)^j$. In particular $\rho_5 = 0$. Consequently

$$r_4(x; M, a) = M^{-6}\Psi(F(x)) + O_a(M^{-8}). \tag{20}$$

2.3. **The constant-field coefficient.** Rotation of the variables and conjugation by the spin lift in $SU(2)$ show that $\Psi(F)$ depends on the vector $B_l = \frac{1}{2}\sum_{\nu,\mu} \varepsilon_{l\nu\mu} F_{\nu\mu}$ only through $|B|$. Take $B = (0,0,b)$, $b \geq 0$, and $A = (-bz_2/2, bz_1/2, 0)$. Set $D_u = \partial + iuA$ and

$$h_u = 1 - \Delta + uW_1 + u^2 W_2, \qquad W_1 = -2iA \cdot \partial + b\sigma_3, \quad W_2 = |A|^2.$$

The Clifford relations give

$$(1 - \gamma \cdot D_u)(1 + \gamma \cdot D_u) = \operatorname{diag}(h_u, h_u), \qquad (1 + \gamma \cdot D_u)^{-1} = (1 - \gamma \cdot D_u)\operatorname{diag}(h_u^{-1}, h_u^{-1}). \tag{21}$$

These are identities of formal series acting on Schwartz spinors; no expansion of an operator with an unbounded connection is being assumed in operator norm.

We compute the fourth coefficient of h_u^{-1} from the kernel

$$k_u(t; x, y) = (4\pi t)^{-3/2} e^{-|x-y|^2/(4t)} \frac{\tau}{\sinh \tau} \exp\left(-\frac{(\tau \coth \tau - 1)|x_p - y_p|^2}{4t} + \frac{iub}{2} x \wedge y \right) e^{-\tau\sigma_3},$$
$$\tau = ubt, \quad x_p = (x_1, x_2), \quad x \wedge y = x_1 y_2 - x_2 y_1. \tag{22}$$

It is analytic in u for $|u|bt < \pi$. The kernel satisfies $(\partial_t + h_u - 1)k_u = 0$. The planar verification uses $\alpha = (ub/4)\coth(ubt)$ and $\phi = -\alpha|x_p - y_p|^2 + iub(x \wedge y)/2$:

$$\frac{\check{h} e^\phi}{e^\phi} = 4\alpha - (4\alpha^2 - u^2 b^2/4)|x_p - y_p|^2, \qquad \alpha' = -(4\alpha^2 - u^2 b^2/4),$$

where $\check{h} = -\partial_1^2 - \partial_2^2 - iub(x_1\partial_2 - x_2\partial_1) + u^2 b^2 |x_p|^2/4$. Differentiation of the prefactor supplies -4α; differentiation of $e^{-\tau\sigma_3}$ supplies the spin term. The longitudinal Gaussian satisfies the one-dimensional heat equation.

Here only the coefficients of (22) are needed. Write $k_u = \sum_{n \geq 0} u^n k^{(n)}$. Each coefficient is the Gaussian times a finite polynomial in $t, x - y, y$, whose monomials $t^j (x-y)^\alpha y^\beta$ have $2j + |\alpha| \geq n$. The factor $e^{iub(x-y)\wedge y/2}$ has weight at least one per power of u, and the other factors have weight at least two, if t has weight two and $x - y$ weight one. Thus

$$K^{(n)}(x, y) = \int_0^\infty e^{-t} k^{(n)}(t; x, y)\, \mathrm{d}t$$

converges off the diagonal and, for $n \geq 1$, on the diagonal. Its terms are y^β times

$$m_{j,\alpha}(x - y) = \int_0^\infty e^{-t} t^j (x-y)^\alpha (4\pi t)^{-3/2} e^{-|x-y|^2/(4t)}\, \mathrm{d}t.$$

Their Fourier transforms are $(i\partial_p)^\alpha[j!(1+|p|^2)^{-j-1}]$, so they define operators preserving Schwartz space.

Coefficientwise integration of the heat equation identifies these operators with the formal inverse of h_u. Indeed, for a Schwartz spinor φ, the zeroth heat coefficient converges to φ as $t \downarrow 0$. The nth coefficient for $n \geq 1$ tends to zero uniformly, since every polynomial term is bounded after integration against φ by $C_\varphi t^{j+|\alpha|/2}$. Integrating from ϵ to infinity and sending ϵ to zero therefore gives

$$(1-\Delta)K^{(n)} + W_1 K^{(n-1)} + W_2 K^{(n-2)} = \delta_{n0} 1.$$

The convergence can be taken in tempered distributions; the polynomial differential operators are continuous there. Since the constant operator $1-\Delta$ is invertible on Schwartz space, the right inverse is the unique formal inverse.

To evaluate the diagonal, the kernel of its fourth coefficient must agree with the continuous kernel already constructed. The functions $m_{j,\alpha}$ are continuous when $2j+|\alpha| \geq 2$ and continuously differentiable when $2j+|\alpha| \geq 3$. For this, write $m_{j,\alpha}(z) = z^\alpha \mu_j(|z|)$ with

$$\mu_j(s) = \int_0^\infty e^{-t} t^j (4\pi t)^{-3/2} e^{-s^2/(4t)} \, \mathrm{d}t.$$

For $j \geq 1$, μ_j is bounded and continuous at zero and $\mu_j'(s) = -s\mu_{j-1}(s)/2$ for $s > 0$; for $j = 0$ use $\mu_0(s) = e^{-s}/(4\pi s)$. These formulas give the asserted regularity at zero. Consequently $K^{(3)}, K^{(4)}$ and $\partial_x K^{(4)}$ are continuous. From (21) the fourth Dirac coefficient has kernel

$$(1 - \gamma\cdot\partial_x)\operatorname{diag}(K^{(4)}, K^{(4)})(x,y) - i\gamma\cdot A(x)\operatorname{diag}(K^{(3)}, K^{(3)})(x,y).$$

It equals $R_A^{(4)}$ because both continuous kernels give the same operator on Schwartz space. At $x = y = 0$, $A(0) = 0$, and the derivative term has zero trace: the two Pauli blocks have opposite signs. It follows that

$$\Psi(F) = 2\int_0^\infty e^{-t} \operatorname{tr} k^{(4)}(t;0,0)\, \mathrm{d}t \tag{23}$$

$$= -\frac{4}{45}(4\pi)^{-3/2}\Gamma(7/2) b^4 = -\frac{1}{48\pi}|B|^4. \tag{24}$$

In the second line we used $\operatorname{tr} k_u(t;0,0) = 2(4\pi t)^{-3/2}\tau\coth\tau$ and $\tau\coth\tau = 1 + \tau^2/3 - \tau^4/45 + O(\tau^6)$.

2.4. **Integration and dilation.** Integrate (20) over the unit neighborhood of $\operatorname{supp} a$, and use (18) on its complement. Equations (15) and (24) give

$$\frac{\mathrm{d}}{\mathrm{d}M} q_{4,M}(a) = -\frac{1}{48\pi M^6}\int |\operatorname{curl} a|^4 \, \mathrm{d}x + O_a(M^{-8}).$$

Integration from M to infinity proves (2).

Finally let $\varphi(x) = e^{-1/(1-|x|^2)}$ for $|x| < 1$ and zero otherwise, set $\beta_{12} = \varphi$, $\beta_{13} = \beta_{23} = 0$, and $a = d^*\beta = (\partial_2\varphi, -\partial_1\varphi, 0)$. The third component of $\operatorname{curl} a$ is $-(\partial_1^2 + \partial_2^2)\varphi$ and has value $4e^{-1}$ at zero. Thus $I = \int |\operatorname{curl} a|^4 \, \mathrm{d}x > 0$. For $\ell \geq 1$ put $\beta^{(\ell)}(x) = \ell^2\beta(x/\ell)$, $a_\ell(x) = \ell a(x/\ell)$, and $f_\ell = dd^*\beta^{(\ell)}$. The unitary dilation $U_\ell\psi(x) = \ell^{-3/2}\psi(x/\ell)$ gives

$$U_\ell^{-1} S_{\bar m} V_{a_\ell} U_\ell = \ell^2 S_{\bar m \ell} V_a, \qquad q_{4,\bar m}(a_\ell) = \ell^8 q_{4,\bar m\ell}(a).$$

For $\bar m \ell \geq 1$, therefore,

$$b(f_\ell) = 24 q_{4,\bar m}(a_\ell) = \frac{\ell^3 I}{10\pi \bar m^5} + O_a(\ell \bar m^{-7}). \tag{25}$$

This is positive for sufficiently large ℓ and completes the proof of theorem A.

3. The chamber symbol and determinant derivatives

The free chamber symbol has a single four-component Dirac singularity. We first isolate it, then use its inverse bounds to control the contact terms in the fourth derivative.

3.1. **The singular part of the inverse.** Use the fiber $V = \ell^2(O_{\mathrm{cub}}; \mathbb{C}^4)$ with measure $48^{-1}\sum_g$ and the constant-label isometry $(Jz)(g) = z$. The orthogonal projection onto its complement is $Q = 1 - JJ^*$. The fiber V has dimension 192, and traces use this full dimension; the label measure does not multiply the matrix trace by $1/48$. Write $\mathcal{D}_m(p)$ for the free symbol of [26a, §2.2], and $R_m(p) = \mathcal{D}_m(p)^{-1}$. For real p it is invertible when $m > 0$.

The light and heavy blocks are the blocks relative to $\operatorname{ran} J \oplus \operatorname{ran} Q$. The heavy inverse, the vanishing of the mixed blocks at zero momentum, and the ellipticity of the light Schur complement are proved in [26a, Proposition 2.2].

We need a bound on the difference between the full inverse and its single light Dirac part, including mixed and heavy blocks. There is $C_\rho < \infty$ such that

$$R_m(p) = J(m + i\gamma \cdot p)^{-1} J^* + \mathcal{E}_m(p), \qquad \|\mathcal{E}_m(p)\| \le C_\rho, \qquad p \in [-\pi, \pi]^3, \quad 0 < m \le 1. \tag{26}$$

To prove this, use the block inversion formula [26a, (16)]. The off-diagonal blocks vanish at zero momentum, and are $O(|p|)$ because only the external hop varies with p. The heavy inverse is bounded by [26a, (10)]. The light Schur complement therefore has the form

$$B_m(p) = m + i\gamma \cdot p + r_m(p), \qquad \|r_m(p)\| \le C_\rho |p|^2.$$

For $|p|$ below a fixed ρ-dependent radius, the product $r_m(p)(m + i\gamma \cdot p)^{-1}$ has norm at most one half. Its Neumann inverse gives

$$\|B_m(p)^{-1} - (m + i\gamma \cdot p)^{-1}\| \le 2C_\rho |p|^2 (m^2 + |p|^2)^{-1} \le 2C_\rho, \qquad |p| \|B_m(p)^{-1}\| \le 2.$$

In the other four terms of the block inverse, each mixed block supplies a factor $|p|$ beside this light inverse. The heavy terms are bounded. This proves (26) near zero. Away from zero, the real-part and high-momentum ellipticity bounds of [26a, §2.2] bound the full inverse uniformly, and the continuum inverse is bounded by $1/|p|$. This proves the estimate on the whole cube. In particular

$$\|R_m(p)\| \le C_\rho + (m^2 + |p|^2)^{-1/2} \le \frac{C'_\rho}{m + |p|}. \tag{27}$$

Define external-hop matrices

$$(T_\nu^{\pm} u)(g) = 1_{\{ge_1 = \pm e_\nu\}} 3(\rho \mp \gamma_\nu) u(g r_{e_1}).$$

Then

$$\begin{aligned} \mathcal{D}_m(p) &= \mathcal{D}_{m,\mathrm{loc}} - \sum_\nu (e^{ip_\nu} T_\nu^+ + e^{-ip_\nu} T_\nu^-), \\ V_\nu(p) &:= \partial_\nu \mathcal{D}_m(p) = -i(e^{ip_\nu} T_\nu^+ - e^{-ip_\nu} T_\nu^-), \\ W_\nu(p) &:= \partial_\nu^2 \mathcal{D}_m(p) = e^{ip_\nu} T_\nu^+ + e^{-ip_\nu} T_\nu^-. \end{aligned} \tag{28}$$

Mixed second derivatives vanish. The disjoint rows indexed by $ge_1 = \pm e_\nu$ imply $\|\sum z_\nu^{\pm} T_\nu^{\pm}\| \le 3(1+\rho) \max |z_\nu^{\pm}|$. Since eight labels contribute to each direction,

$$J^* T_\nu^{\pm} J = \tfrac{1}{2}(\rho \mp \gamma_\nu), \qquad J^* V_\nu(0) J = i\gamma_\nu. \tag{29}$$

All nonzero derivatives of the symbol are bounded by six on real momenta. Differentiating $R_m \mathcal{D}_m = 1$ gives

$$\partial^\alpha R_m = - \sum_{0 < \delta \le \alpha} \binom{\alpha}{\delta} (\partial^{\alpha - \delta} R_m)(\partial^\delta \mathcal{D}_m) R_m.$$

Starting with (27), induction bounds each summand by a constant times $(\operatorname{dist}(p, 2\pi\mathbb{Z}^3) + m)^{-1-|\alpha|}$; the remaining factor of the distance has nonnegative exponent and is bounded by a power of seven. Thus

$$\|\partial^\alpha R_m(p)\| \le C_{\alpha,\rho}(\operatorname{dist}(p, 2\pi\mathbb{Z}^3) + m)^{-1-|\alpha|}. \tag{30}$$

The involution $(\mathcal{U}u)(g) = \gamma_5 u(-g)$ satisfies

$$\mathcal{U}\mathcal{D}_m(p)\mathcal{U} = \mathcal{D}_m(-p), \qquad \mathcal{U}V_\nu(p)\mathcal{U} = -V_\nu(-p). \tag{31}$$

Indeed $-g$ reverses every bond direction, and $\gamma_5\gamma_\nu\gamma_5 = -\gamma_\nu$ restores the hopping matrix. The external phase then changes p to $-p$. This parity will cancel the first momentum derivative of the polarization kernel.

3.2. **Derivative and contact terms.** Let $\mathcal{P}_n = \frac{2\pi}{n}(\mathbb{Z} + \frac{1}{2})^3 \cap (-\pi, \pi]^3$. The normalized Bloch vectors are $n^{-3/2}e^{ip\cdot c}$, and the lattice free propagator G is the multiplier $R_m(p)$ on these antiperiodic momenta. Write $A_\nu(c) = a_{\eta,\nu}(\eta c)$ and $A_\nu^-(c) = A_\nu(c - e_\nu)$, and let $U_\nu^\pm\psi(c) = \psi(c \pm e_\nu)$. The bond values are $b_\nu^+ = \eta A_\nu$, $b_\nu^- = -\eta A_\nu^-$. The exact connection derivatives are

$$D_j = -\sum_{\nu,\pm}(ib_\nu^\pm)^j T_\nu^\pm U_\nu^\pm, \quad D^{\mathrm{ch}}[B_t] - D^{\mathrm{ch}}[0] = \sum_{j\ge1}\frac{t^j}{j!}D_j, \quad Y_j = GD_j. \tag{32}$$

The shifts act first. These D_j are derivatives, with the factorials outside the vertices.

For small complex t, $N_0(t) = \det(1+\sum_{j\ge1} t^j Y_j/j!)$ and its zero-at-zero logarithm is the convergent finite-matrix trace series $\sum_{l\ge1}(-1)^{l+1}\operatorname{Tr}(\sum_j t^j Y_j/j!)^l/l$. The tail $l \ge 5$ is $O(t^5)$, so it contributes no fourth derivative. Cyclicity of the trace in the terms $l \le 4$ gives

$$\begin{aligned} b_{\mathrm{ch}} = {} & \operatorname{Tr} Y_4 - 4\operatorname{Tr}(Y_1Y_3) - 3\operatorname{Tr} Y_2^2 \\ & + 12\operatorname{Tr}(Y_1^2Y_2) - 6\operatorname{Tr} Y_1^4. \end{aligned} \tag{33}$$

For example, the cyclically equal orders $(1,1,2)$, $(1,2,1)$ and $(2,1,1)$ contribute $24\cdot\frac{1}{3}\cdot\frac{3}{2} = 12$. The quadratic identity is $a_{\mathrm{ch}} = \operatorname{Tr} Y_2 - \operatorname{Tr} Y_1^2$. The determinant series is used only near zero; the determinant itself is entire in t.

We bound all four contact traces in (33) uniformly in R and the position of the probe. The interpolating periodic field is

$$\widetilde{A}_\nu(y) = \sum_z a^0_{\eta,\nu}(y + \eta e_\nu/2 + Rz), \qquad A_\nu(c) = \widetilde{A}_\nu(\eta c).$$

When $R \ge D_\beta + 1$, averaging translates of the derivatives of β shows

$$\begin{gathered} \|\widetilde{A}\|_\infty + \|\partial\widetilde{A}\|_\infty \le C_\beta^{\mathrm{free}}, \qquad \|\widetilde{A} - a^R\|_\infty \le C_\beta^{\mathrm{free}}\eta, \\ \eta^3|\{c : A_\nu(c) \ne 0 \text{ for some } \nu\}| \le (D_\beta+3)^3, \qquad \|\eta A_\nu\|_{\ell^4} \le C_\beta^{\mathrm{free}}\eta^{1/4}, \\ \|\eta(A_\nu^- - A_\nu)\|_{\ell^4} \le C_\beta^{\mathrm{free}}\eta^{5/4}, \qquad \|\widetilde{A} - a^R\|_{L^4(T_R)} \le C_\beta^{\mathrm{free}}\eta. \end{gathered} \tag{34}$$

Every averaging displacement has size at most η in sup norm. The translates of $\operatorname{supp}\beta$ are disjoint, so their derivative sup norms are unchanged. The supports of the interpolants lie in $Q_\beta + [-\eta,\eta]^3$ modulo the periods. Its volume and number of lattice points give the last estimates, including the union with its one-step translate for the difference.

Let $\widehat{A_\nu^j}(k) = n^{-3}\sum_c A_\nu(c)^j e^{-ik\cdot c}$ on the finite momentum group $\mathcal{G}_n = \frac{2\pi}{n}\mathbb{Z}^3/2\pi\mathbb{Z}^3$, and set $F_j(k) = \sum_\nu |\widehat{A_\nu^j}(k)|$. There are constants $C_{\mathrm{Four},1}, C_{\mathrm{Four},0}$ depending only on β with

$$\|F_j\|_{\ell^1(\mathcal{G}_n)} \le C_{\mathrm{Four},1}^j, \qquad R^3\|F_j\|_\infty \le C_{\mathrm{Four},1}^{j-1}C_{\mathrm{Four},0}. \tag{35}$$

The transform of the centered difference and the cube average in (3) is

$$-\sum_{\mu\ne\nu} iP_\mu \operatorname{sinc}(\eta P_\mu/2)\prod_l \operatorname{sinc}(\eta P_l/2)\widehat{\beta}_{\mu\nu}(P),$$

where $\operatorname{sinc}(z) = \sin z/z$ and $\operatorname{sinc}(0) = 1$. The midpoint shift multiplies this transform by a phase of modulus one. The shifted transform has modulus at most $C_\beta^{\mathrm{free}}(1+|P|)^{-4}$. Poisson summation [Gut15, Theorems 4–7] gives

$$\widehat{A}_\nu(k) = R^{-3} \sum_{z\in\mathbb{Z}^3} a^0_{\eta,\nu}(\widehat{\cdot + \eta e_\nu}/2)((k+2\pi z)/\eta).$$

Summing over representatives k and z gives a single sum over $\frac{2\pi}{R}\mathbb{Z}^3$, bounded after multiplication by R^{-3} by its integrable radial majorant, uniformly for $R \geq 1$. For $|k|_\infty \leq \pi$, nonzero z have $|(k+2\pi z)/\eta| \geq \pi|z|_\infty/\eta$; the convergent sum $\sum_{z\neq 0}|z|_\infty^{-4}$ gives the sup bound. Powers of A_ν give convolutions on $\mathcal{G}_n$; $\ell^1 * \ell^1 \to \ell^1$ and $\ell^1 * \ell^\infty \to \ell^\infty$ give (35) for all j.

The momentum blocks of a vertex are

$$D_j(p,p') = -\eta^j \sum_\nu \widehat{A^j_\nu}(p-p')[i^j e^{ip'_\nu} T^+_\nu + (-i)^j e^{-ip_\nu} T^-_\nu], \qquad \|D_j(p,p')\| \leq 6\eta^j F_j(p-p').$$

For $l \leq 3$ and $j_1 + \cdots + j_l = 4$, apply these bounds and (27) to the cyclic trace. With differences $k_i = p_i - p_{i+1}$, the sum is parametrized by $p_1, k_1, \ldots, k_{l-1}$. Hölder in p_1 bounds the product of its l translates of $(m+|p|)^{-1}$ by $\sum_p (m+|p|)^{-l}$. Bound the final F_{j_l} in sup norm and the others in ℓ^1. Thus

$$|\operatorname{Tr}(Y_{j_1}\cdots Y_{j_l})| \leq 192(6C'_\rho)^l C^3_{\mathrm{Four},1} C_{\mathrm{Four},0}\,\eta \left(n^{-3}\sum_{p\in\mathcal{P}_n}(m+|p|)^{-l}\right). \tag{36}$$

The factor η is $\eta^4 n^3/R^3$.

For antiperiodic momenta, comparison with their disjoint cubes of side $2\pi/n$ gives, for a decreasing radial function h, $n^{-3}\sum_p h(|p|) \leq \pi^{-3}\int h(|y|)\,\mathrm{d}y$. Indeed in the cube of p one has $|y| \leq 2|p|$, and a rescaling by two gives the factor π^{-3}. On the bounded momentum cube this yields constants for $l = 1, 2$ and $\log(1+5/m)$ for $l = 3$. Inserting the multiplicities $1, 4, 3, 12$ from (33) proves

$$\begin{aligned} &|\operatorname{Tr} Y_4| + 4|\operatorname{Tr}(Y_1 Y_3)| + 3|\operatorname{Tr} Y_2^2| + 12|\operatorname{Tr}(Y_1^2 Y_2)| \\ &\qquad \leq C_{\mathrm{ct}}(\rho,\beta)\,\eta\,(1+\log(1+5/(\bar{m}\eta)))\,. \end{aligned} \tag{37}$$

The constant does not depend on $\bar{m}$, η, R or the position of $\operatorname{supp}\beta$.

4. Comparison of the fourth traces

The contact bound reduces the first estimate in (8) to the trace with four first-order vertices. We compare that trace with the continuum torus trace in five steps. The Schatten-four norm controls each replacement, and the Yukawa kernel controls the remaining volume error.

Proof of the fourth-coefficient assertions in theorem B. We first establish the trace bounds used in the comparisons. For a lattice multiplier M_h with fiber symbol $h(p)$ and a scalar multiplication operator v, one has

$$\|M_h v\|_4 \leq \left(n^{-3}\sum_{p\in\mathcal{P}_n}\|h(p)\|_4^4\right)^{1/4}\|v\|_{\ell^4}. \tag{38}$$

To prove it put $U = M_{h^*h}$ and $V = |v|^2$. Expanding the nonnegative trace of $(UV-VU)^*(UV-VU)$ gives $\operatorname{Tr}(UVUV) \leq \operatorname{Tr}(U^2V^2)$. The left side is $\|M_h v\|_4^4$, and each normalized Bloch vector has constant squared modulus n^{-3}. Evaluating the right side in that basis gives (38). On the continuum torus the same proof, first for a finite set of momenta, gives

$$\|M_h v\|_4 \leq \left(R^{-3}\sum_{P\in\mathcal{P}_R}\|h(P)\|_4^4\right)^{1/4}\|v\|_{L^4(T_R)}. \tag{39}$$

Truncation in momentum converges in the Schatten norm when the sum is finite, and its operator-norm limit is $M_h v$, proving the general form.

The cube comparison used above also gives

$$\begin{aligned} n^{-3}\sum_{p\in\mathcal{P}_n}(m^2+|p|^2)^{-2} &\le \frac{1}{\pi m}, \qquad R^{-3}\sum_{P\in\mathcal{P}_R}(\bar m^2+|P|^2)^{-2} \le \frac{1}{\pi \bar m},\\ R^{-3}\sum_{|P|_\infty\le \pi/(2\eta)}|P|^{-2} &\le C\eta^{-1}, \qquad R^{-3}\sum_{|P|_\infty>\pi/(2\eta)}|P|^{-4} \le C\eta. \end{aligned} \tag{40}$$

For the last bound use the decreasing radial majorant $\min((2\eta/\pi)^4, |P|^{-4})$; its integral is $O(\eta)$. The preceding bounds and $\|s_{\bar m}(P)\|_4^4 = 4(\bar m^2+|P|^2)^{-2}$ show that $S^R i\gamma\cdot a^R$ has a bounded Schatten-four norm, uniformly in $R \ge D_\beta + 1$. The same estimate applies to the interpolating field $\widetilde{A}$.

On $\mathbb{R}^3$, let

$$Y_{\bar m}(x) = \frac{e^{-\bar m|x|}}{4\pi|x|}, \qquad S_{\bar m}(x) = (\bar m - \gamma\cdot\partial)Y_{\bar m}(x).$$

The scalar Fourier transform is $(\bar m^2+|P|^2)^{-1}$; the derivative has no point mass, since its boundary term on a sphere of radius r is $O(r)$. Its Fourier transform therefore gives $s_{\bar m}(P)$. These kernels are integrable, so their convolutions are the corresponding bounded Fourier multipliers on L^2. Also $\|Y_{\bar m}\|_{L^2}^2 = (8\pi\bar m)^{-1}$. For bounded compactly supported v, the kernel of $v^* S_{\bar m}^* S_{\bar m} v$ is $\overline{v(x)}Y_{\bar m}(x-y)v(y)1_4$. Tonelli and Cauchy–Schwarz in the translated factors give

$$\|S_{\bar m} v\|_4^4 \le 4\|Y_{\bar m}\|_{L^2}^2\|v\|_{L^4}^4 = \frac{\|v\|_{L^4}^4}{2\pi\bar m}.$$

For the matrix multiplication $i\gamma\cdot a$, the Clifford identity gives the same proof with $|a|$ in place of $|v|$. Thus

$$\|S_{\bar m} i\gamma\cdot a\|_4^4 \le (2\pi\bar m)^{-1}\int_{\mathbb{R}^3}|a|^4\,\mathrm{d}x. \tag{41}$$

The traces in (7) exist by Schatten Hölder.

To check reality, treat $S = S_{\bar m}$ or S^R and $v = i\gamma\cdot a$ or $i\gamma\cdot a^R$ together. Factor $v = |a|^{1/2}(i\gamma\cdot a/|a|)|a|^{1/2}$, with the middle factor zero at its zeros. The preceding estimates put SvS in the Hilbert–Schmidt class. One has $\gamma_5 S\gamma_5 = S^*$ and $\gamma_5 v\gamma_5 = v^*$. Hence conjugating $\operatorname{Tr}[(SvS)v(SvS)v]$ by γ_5 gives its complex conjugate after cyclically moving the bounded v past a trace-class product of the two Hilbert–Schmidt factors. This proves reality.

We now compare the lattice fourth-power trace with its continuum torus value. The four-power inequality used throughout is

$$|\operatorname{Tr}X^4 - \operatorname{Tr}Z^4| \le 4\max(\|X\|_4,\|Z\|_4)^3\|X-Z\|_4. \tag{42}$$

It follows by expanding $X^4 - Z^4$ into four products with one $X - Z$ factor and applying Schatten Hölder. All the operators compared below have uniformly bounded Schatten-four norms; each bound follows from (38) or (39). In particular (27) gives $n^{-3}\sum_p \|R_m(p)\|_4^4 \le C_\rho(1+m^{-1})$, which, with $\|\eta A_\nu\|_{\ell^4} \le C_\beta^{\mathrm{free}}\eta^{1/4}$, proves $\|Y_1\|_4 \le C_{\rho,\bar m,\beta}$.

Removing the bounded part of the inverse. Let $\chi(p) = 1_{\{|p|_\infty\le\pi/2\}}$ and S_c be the lattice multiplier $\chi(p)s_m(p)$ on four-component antiperiodic spinors. The complementary inverse $R_m(p) - J\chi(p)s_m(p)J^*$ is uniformly bounded: (26) controls the whole cube, and s_m is bounded on $1-\chi$. Thus

$$\|Y_1 - JS_cJ^*D_1\|_4 \le C_{\rho,\beta}\eta^{1/4}.$$

The two norms in (42) are bounded by the same estimates. Since $J^*J = 1$, cyclicity reduces the fourth-power trace of the light operator to that of $S_c w_1$, where

$$w_1 = J^*D_1J = -\frac{i\eta}{2}\sum_\nu[A_\nu(\rho-\gamma_\nu)U_\nu^+ - A_\nu^-(\rho+\gamma_\nu)U_\nu^-]. \tag{43}$$

This comparison costs $C\eta^{1/4}$ in the trace. The compression uses (29), and therefore has one four-component light species.

Replacing the vertex by multiplication. Move shifts to the left by $A_\nu U_\nu^+ = U_\nu^+ A_\nu^-$ and $A_\nu^- U_\nu^- = U_\nu^- A_\nu$. Then

$$w_1 - i\eta\gamma\cdot A = -\frac{i\eta}{2}\sum_\nu \big[(\rho-\gamma_\nu)(U_\nu^+ - 1)A_\nu^- - (\rho+\gamma_\nu)(U_\nu^- - 1)A_\nu \\ + (\rho-\gamma_\nu)(A_\nu^- - A_\nu)\big]. \tag{44}$$

The multiplier $\chi(p)s_m(p)(\rho \mp \gamma_\nu)(e^{\pm ip_\nu} - 1)$ has norm at most two, since $|e^{ip_\nu} - 1| \le |p_\nu|$. Its Schatten-four multiplier average is consequently bounded independently of the mass. The first two terms of (44) then cost $O(\eta^{1/4})$. The last term costs $O(m^{-1/4}\eta^{5/4}) = O(\eta)$ by (34). Therefore $\|S_c(w_1 - i\eta\gamma\cdot A)\|_4 \le C\eta^{1/4}$, and (42) gives the same trace error.

Identifying the truncated continuum momenta. The map $p \mapsto P = p/\eta$ identifies the retained lattice momenta with $\mathcal{P}_R^{\le} = \{P \in \mathcal{P}_R : |P|_\infty \le \pi/(2\eta)\}$, and $\eta s_m(p) = s_{\bar m}(P)$. The operator $S_c i\eta\gamma\cdot A$ has range in this truncated space. Its fourth trace is thus that of its compression to the space. In the normalized momentum basis this compression has blocks

$$T_{\mathrm{lat}}(P,P') = s_{\bar m}(P) i\sum_\nu \gamma_\nu n^{-3}\sum_c A_\nu(c)e^{-i\eta(P-P')\cdot c}.$$

Let $\widetilde{A}$ be the smooth interpolant and $\widehat{A}_R(k) = R^{-3}\int_{T_R}\widetilde{A}(y)e^{-ik\cdot y}\,\mathrm{d}y$. Absolute Fourier convergence, using four derivatives of the interpolant, gives the exact alias identity

$$n^{-3}\sum_c A_\nu(c)e^{-i\eta k\cdot c} = \widehat{A}_{R,\nu}(k) + \sum_{z\neq 0}\widehat{A}_{R,\nu}(k + 2\pi z/\eta). \tag{45}$$

For $k = P - P'$, $|k|_\infty \le \pi/\eta$. Four integrations by parts give $|\widehat{A}_{R,\nu}(k')| \le C_\beta^{\mathrm{free}} R^{-3}|k'|_\infty^{-4}$; the nonzero aliases in (45) are therefore bounded by $C_\beta^{\mathrm{free}} R^{-3}\eta^4$.

Replacing the discrete matrix elements by $\widehat{A}_R$ gives the compression T_{int} of $\chi S^R i\gamma\cdot\widetilde{A}$. Its norm is bounded. For the difference, sum the squared Hilbert–Schmidt norms of its blocks. There are at most $R^3\eta^{-3}$ choices of P', while $\sum_{P\in\mathcal{P}_R^{\le}}|P|^{-2} \le CR^3\eta^{-1}$. Since $\|\gamma\cdot z\|_2^2 = 4|z|^2$, the alias bound gives

$$\|T_{\mathrm{lat}} - T_{\mathrm{int}}\|_2^2 \le C_\beta^{\mathrm{free}}(R^3\eta^{-1})(R^3\eta^{-3})(R^{-3}\eta^4)^2 = C_\beta^{\mathrm{free}}\eta^4.$$

All powers of R cancel. The Schatten-four norm is at most the Hilbert–Schmidt norm, so the trace error in this comparison is $O(\eta^2)$.

Replacing the interpolant and removing the cutoff. The difference between the compressions of $\chi S^R i\gamma\cdot\widetilde{A}$ and $\chi S^R i\gamma\cdot a^R$ has Schatten-four norm $O(\eta)$ by (39) and (34). For an operator X with $X = \chi X$, cyclicity gives $\operatorname{Tr} X^4 = \operatorname{Tr}(\chi X\chi)^4$; the same holds for $X = X\chi$. Thus we can remove the compression and compare $\chi S^R i\gamma\cdot a^R$ directly with $S^R i\gamma\cdot a^R$. The removed momentum tail in (40) gives

$$\|(1-\chi)S^R i\gamma\cdot a^R\|_4 \le C_\beta^{\mathrm{free}}\eta^{1/4}.$$

These are respectively the fourth and fifth comparisons; their trace errors are $O(\eta)$ and $O(\eta^{1/4})$. The preceding three comparisons, together with (37), prove the first line of (8). Here $\eta(1+\log(1+5/(\bar m\eta))) \le C_{\bar m}\eta^{1/4}$.

It remains to compare torus and infinite-space traces. Because the multiplication by a is supported in Q_β, both traces can be computed on $L^2(Q_\beta;\mathbb{C}^4)$, using extension by zero to $\mathbb{R}^3$ or antiperiodic extension to T_R. The same compression identity proves this localization. Convolution by the integrable kernel $S_{\bar m}(x)$ on antiperiodic functions agrees with S^R on its momentum basis and hence

on L^2. After restricting its two variables to Q_β, its difference from the infinite-space kernel is

$$W_R(x,y) = \sum_{z\in\mathbb{Z}^3\setminus\{0\}} (-1)^{z_1+z_2+z_3} S_{\bar m}(x-y-Rz).$$

For $|u| \geq 1$ the Yukawa formula gives $\|S_{\bar m}(u)\| \leq (3/(4\pi))e^{-\bar m|u|}$. Moreover $|x-y-Rz| \geq R|z|_\infty - D_\beta \geq (R-D_\beta) + |z|_\infty - 1 \geq 1$. Counting $24j^2+2$ images at $|z|_\infty = j$ yields

$$\|W_R(x,y)\| \leq C_{\bar m} e^{-\bar m(R-D_\beta)}, \qquad x,y \in Q_\beta.$$

The series converges absolutely and uniformly there. Multiplication by $i\gamma\cdot a(y)$ then gives a Hilbert–Schmidt difference with this same exponential bound, since Q_β has finite volume. The infinite localized operator has bounded Schatten-four norm, and the torus one is bounded by it and the difference. Applying (42) proves the second line of (8).

The constants above involve only ρ, $\bar m$, the side of the support cube, derivative sup and L^1 norms of β, and moduli of its Fourier transform. All are unchanged by a real translation. This gives the asserted uniformity and the limit. □

5. Symmetry and the second-order response

5.1. **Evenness and placement.** The coefficients in (6) are real, and

$$c_{2j+1} = 0, \qquad c_2 = a_{\mathrm{ch}}, \qquad c_4 = b_{\mathrm{ch}} + 3c_2^2. \tag{46}$$

We prove the determinant identities for complex external connections to obtain evenness for every complex value of t.

For a real connection, let $(R_i(B)\psi)(c,g) = \psi(c, gs_i)$ for $i = 0,1,2$, and

$$(R_3(B)\psi)(c,g) = e^{iB(c,c+ge_1)}\psi(c+ge_1, gr_{e_1}).$$

The outgoing unit bond direction is $\widehat{d}_i(c,g) = -gn_i$. Each $R_i(B)$ is self-adjoint and unitary. The direction of a bond changes sign at its other endpoint. Consequently

$$D^{\mathrm{ch}}[B] = \frac{\rho}{2}\sum_i w_i(1-R_i(B)) + \frac{1}{2}\sum_i w_i(\gamma\cdot\widehat{d}_i)R_i(B),$$

where the first sum is nonnegative and the second is anti-self-adjoint. The two spin sectors $\gamma_\mu = \mathrm{diag}(\sigma_\mu, -\sigma_\mu)$ give adjoint restrictions, as in [26b, (17)]. Thus $\det(D^{\mathrm{ch}}[B]+m)$ is a squared absolute value. The real part of $D^{\mathrm{ch}}[B]+m$ is at least $m>0$, so the determinant is strictly positive. In particular $N_0(t) > 0$ for real t.

The transpose identity extends charge symmetry to complex connections. Let $C = \sigma_2 \oplus \sigma_2$ be the charge-conjugation matrix of [26b, §3.1]. Transposing a phased bond reverses its direction and changes its phase from B to $-B$. The identity $C\gamma_v^\top C^{-1} = -\gamma_v$ changes the transposed matrix at the other endpoint back to the matrix at the original one. The diagonal part is scalar; hence

$$C(D^{\mathrm{ch}}[B]+m)^\top C^{-1} = D^{\mathrm{ch}}[-B] + m.$$

The transpose here is for the bilinear chamber pairing. In the site basis this pairing is $1/48$ times the identity, so this is the ordinary matrix transpose and preserves the determinant. Every matrix entry is a finite sum of constants and exponentials of linear functions of t. Therefore N_0 is entire, and the transpose identity gives $N_0(-t) = N_0(t)$ for all $t \in \mathbb{C}$. Expanding its logarithm at zero proves (46).

We also record exact covariance under lattice translations and cubic reflections. Let $g = (v,r)$, where $v \in \mathbb{Z}^3$ and r is a signed permutation. A reflection with normal $\widehat{n}$ has the spin lift $S_{\widehat{n}} = (\gamma\cdot\widehat{n})\gamma_5$, for which

$$S_{\widehat{n}}(\gamma\cdot z)S_{\widehat{n}}^{-1} = \gamma\cdot(z - 2(z\cdot\widehat{n})\widehat{n}), \qquad z\in\mathbb{C}^3.$$

Choose a word representing r in the three linear chamber reflections and let S_g be the corresponding product. It follows that $S_g(\gamma\cdot z)S_g^{-1}=\gamma\cdot rz$. The site map $(c,h)\mapsto(rc+v,rh)$ commutes with each bond involution and carries $\widehat{d}_i(c,h)$ to $r\widehat{d}_i(c,h)$. Therefore, for

$$(U_g\psi)(x)=S_g\psi(g^{-1}x),\qquad (g_*B)(c,c')=B(g^{-1}c,g^{-1}c'),$$

the hopping formula gives

$$D^{\mathrm{ch}}[g_*B]+m=U_g(D^{\mathrm{ch}}[B]+m)U_g^{-1},\qquad B\text{ complex.}\tag{47}$$

This identity descends to the antiperiodic torus. Indeed $r^{-1}e_\mu=\pm e_\nu$, and either translation by ne_ν or by $-ne_\nu$ changes an antiperiodic spinor's sign. Thus U_g and its inverse preserve the spinor space. Taking determinants in (47) shows that N_0, c_j, a_{ch} and b_{ch} are unchanged when the external cochain is replaced by g_*B_t.

For a physical translation $v\in\eta\mathbb{Z}^3$, cube averaging and centered differences commute with translation. Periodization in (4) then gives $a_\eta[\beta(\cdot-v)](y)=a_\eta[\beta](y-v)$. This is exactly the cochain action with lattice translation v/η. The same determinant coefficients can therefore be used at each lattice placement of the probe and at each of its reflected images.

5.2. **The two-photon coefficient.** The coefficient is

$$a_{\mathrm{ch}}=\mathrm{Tr}(GD_2)-\mathrm{Tr}(GD_1GD_1).$$

Estimating the two traces separately would lose a power of the mesh. We keep their cancellation by expressing their difference as a polarization kernel. Its integral vanishes at zero momentum; parity removes the first momentum derivative.

Define the bosonic momentum representatives

$$\mathcal{K}_{\eta,R}=\left\{\frac{2\pi}{R}j:j\in\mathbb{Z}^3,\ -n/2<j_\nu\le n/2\right\},\qquad n=R/\eta,$$

and the midpoint transform of the unperiodized probe

$$h_{\eta,\nu}(K)=\eta^3\sum_{y\in\eta\mathbb{Z}^3+\eta e_\nu/2}a^0_{\eta,\nu}(y)e^{-iK\cdot y}.\tag{48}$$

The sum is finite. For real p,k put

$$\begin{aligned}L_{\mu\nu}(p,k)&=\delta_{\mu\nu}\,\mathrm{tr}[R_m(p)W_\nu(p)]\\&\quad-\mathrm{tr}[R_m(p)V_\mu(p+k/2)R_m(p+k)V_\nu(p+k/2)],\end{aligned}\tag{49}$$

$$\Pi^{\eta,R}_{\mu\nu}(K)=\eta^{-1}n^{-3}\sum_{p\in\mathcal{P}_n}L_{\mu\nu}(p,\eta K),$$

$$\Pi^{\eta}_{\mu\nu}(K)=\eta^{-1}\int_{[-\pi,\pi]^3}L_{\mu\nu}(p,\eta K)\frac{\mathrm{d}p}{(2\pi)^3}.\tag{50}$$

Only $\Pi^{\eta,R}$ is a finite-volume polarization; the integral Π^η retains the lattice symbol but has no volume parameter. The exact coefficient is

$$a_{\mathrm{ch}}(\eta,R)=R^{-3}\sum_{K\in\mathcal{K}_{\eta,R}}\sum_{\mu,\nu}\overline{h_{\eta,\mu}(K)}h_{\eta,\nu}(K)\Pi^{\eta,R}_{\mu\nu}(K).\tag{51}$$

To check the midpoint and normalization in this formula, write $b_\nu(c) = \eta A_\nu(c)$ and $\widehat{b}_\nu(k) = n^{-3}\sum_c b_\nu(c)e^{-ik\cdot c}$. The derivative vertices of section 3.2 give the blocks

$$D_1[p+k,p] = \sum_\nu \widehat{b}_\nu(k)e^{-ik_\nu/2}V_\nu(p+k/2),$$
$$D_1[p,p+k] = \sum_\mu \overline{\widehat{b}_\mu(k)e^{-ik_\mu/2}}V_\mu(p+k/2),$$
$$D_2[p,p] = \sum_\nu \sum_k |\widehat{b}_\nu(k)|^2 W_\nu(p).$$

Here k ranges over $\eta\mathcal{K}_{\eta,R}$; the last identity uses Parseval for the real function b_ν. For example, the positive hop in $D_1[p+k,p]$ is $-i\widehat{b}_\nu(k)e^{ip_\nu}T_\nu^+$, and the negative hop is $i\widehat{b}_\nu(k)e^{-i(p_\nu+k_\nu)}T_\nu^-$. Their sum is the stated midpoint vertex. Under $k_\nu \mapsto k_\nu + 2\pi$ both the phase and that vertex change sign, so their product is well-defined on momentum representatives.

Unfolding the periodization gives, for $k = \eta K$,

$$\widehat{b}_\nu(k)e^{-ik_\nu/2} = n^{-3}\eta \sum_{c\in\mathbb{Z}^3} a^0_{\eta,\nu}(\eta c + \eta e_\nu/2)e^{-iK\cdot(\eta c+\eta e_\nu/2)} = \eta R^{-3} h_{\eta,\nu}(K).$$

Substitute these three block identities into the two traces. The scalar normalization is $\eta^2 R^{-6} = R^{-3}\eta^{-1}n^{-3}$, which proves (51). This identity requires no separation of the probe from its periodic images.

Proposition 5.1. There is $C_\rho < \infty$, independent of $0 < \bar{m} \le 1$, $0 < \eta \le 1$, $R/\eta \in \mathbb{N}$, $R/\eta \ge 2$, and $K \in \mathbb{R}^3$, such that

$$|\Pi^{\eta,R}_{\mu\nu}(K) - \Pi^\eta_{\mu\nu}(K)| \le C_\rho \bar{m}^{-3}R^{-4}, \tag{52}$$
$$|\Pi^\eta_{\mu\nu}(K)| \le C_\rho \bar{m}^{-1}|K|^2, \qquad |\nabla_K \Pi^\eta_{\mu\nu}(K)| \le C_\rho \bar{m}^{-1}|K|. \tag{53}$$

■

Proof. We first estimate the quadrature error. If F is a C^4 periodic function on the momentum torus, its Fourier coefficients satisfy

$$|\widehat{F}(z)| \le |z|_\infty^{-4} \max_j \int_{[-\pi,\pi]^3} |\partial_j^4 F(p)| \frac{\mathrm{d}p}{(2\pi)^3}, \qquad z \ne 0.$$

This follows by four integrations by parts in a coordinate attaining $|z|_\infty$. The coefficients are absolutely summable, since $\sum_{z\ne 0}|z|_\infty^{-4} < 42$. Their Fourier series equals F: its difference from F has zero Fourier coefficients, and density of trigonometric polynomials in continuous periodic functions makes the difference zero. The antiperiodic mean selects the coefficients at nz with sign $(-1)^{z_1+z_2+z_3}$. Consequently

$$\left| n^{-3}\sum_{p\in\mathcal{P}_n} F(p) - \int_{[-\pi,\pi]^3} F(p)\frac{\mathrm{d}p}{(2\pi)^3}\right| \le 42n^{-4}\max_j \int_{[-\pi,\pi]^3} |\partial_j^4 F(p)|\frac{\mathrm{d}p}{(2\pi)^3}. \tag{54}$$

Set $x(p) = \mathrm{dist}(p, 2\pi\mathbb{Z}^3) + m$. On a period cube $x(p) = |p| + m$ and $x(p) < 7$. The inverse derivative estimate (30), used up to order four, and the bounded symbol derivatives give

$$|\partial_j^4 L_{\mu\nu}(p,k)| \le C_\rho[x(p)^{-6} + x(p+k)^{-6}]. \tag{55}$$

For the contact term, each Leibniz summand has at most five powers of $x(p)^{-1}$; the remaining power is supplied by $x(p) < 7$. For the bubble term, a summand with a derivatives on the first inverse and b on the second has the factor $x(p)^{-1-a}x(p+k)^{-1-b}$, with $a+b \le 4$. The other derivatives fall on bounded vertices. Since $u^{-s}v^{-t} \le u^{-s-t} + v^{-s-t}$ for $s,t > 0$, and $x < 7$, the factor is bounded by a constant times the right side of (55). This bound is uniform in the shift k.

For $j > 3$, periodicity and the radial integral give

$$\int_{[-\pi,\pi]^3} x(p+u)^{-j}\,\mathrm{d}p \le 4\pi \int_0^\infty \frac{r^2\,\mathrm{d}r}{(r+m)^j} = \frac{8\pi m^{3-j}}{(j-1)(j-2)(j-3)}, \qquad u \in \mathbb{R}^3. \tag{56}$$

Indeed the integral over a translated period cube is unchanged, and on the original cube it is bounded by the integral over $\mathbb{R}^3$. The case $j = 6$ bounds the integral in (54) by $C_\rho m^{-3}$. The definitions of the two polarizations then give the first estimate, because

$$\eta^{-1} n^{-4} m^{-3} = \eta^{-1}(\eta/R)^4(\bar{m}\eta)^{-3} = \bar{m}^{-3} R^{-4}.$$

For the quadratic bound, use the symmetric bubble

$$F_{\mu\nu}(p,k) = \operatorname{tr}[R_m(p-k/2)V_\mu(p)R_m(p+k/2)V_\nu(p)].$$

It is periodic in p. Translating the period cube by $-k/2$ shows that its integral is the integral of the bubble in (49). At $k = 0$ the whole integrand is a total derivative:

$$L_{\mu\nu}(p,0) = \partial_\mu \operatorname{tr}[R_m(p)V_\nu(p)]. \tag{57}$$

In fact $\partial_\mu R_m = -R_m V_\mu R_m$ and $\partial_\mu V_\nu = \delta_{\mu\nu} W_\nu$. Its integral therefore vanishes, including the contact term. We obtain

$$\Pi^\eta_{\mu\nu}(K) = -\eta^{-1} \int_{[-\pi,\pi]^3} [F_{\mu\nu}(p,\eta K) - F_{\mu\nu}(p,0)] \frac{\mathrm{d}p}{(2\pi)^3}. \tag{58}$$

Conjugation by the involution in (31) changes (p,k) to $(-p,-k)$ in $F_{\mu\nu}$; the signs of its two vertices cancel. Thus $F_{\mu\nu}(-p,-k) = F_{\mu\nu}(p,k)$. Its first k derivative at zero is odd in p, and hence

$$\int_{[-\pi,\pi]^3} (\partial_{k_a} F_{\mu\nu})(p,0)\,\mathrm{d}p = 0. \tag{59}$$

This removes the linear term as well as the constant term removed by (57).

The Hessian has two derivatives in total on the inverse factors; the midpoint vertices no longer depend on k. With $p_\pm = p \pm k/2$ and the vertices evaluated at p, it is

$$\begin{aligned}\partial_{k_a}\partial_{k_b} F_{\mu\nu} = \tfrac{1}{4}\big(&\operatorname{tr}[(\partial_a\partial_b R_m)(p_-)V_\mu R_m(p_+)V_\nu] \\ &- \operatorname{tr}[(\partial_a R_m)(p_-)V_\mu(\partial_b R_m)(p_+)V_\nu] \\ &- \operatorname{tr}[(\partial_b R_m)(p_-)V_\mu(\partial_a R_m)(p_+)V_\nu] \\ &+ \operatorname{tr}[R_m(p_-)V_\mu(\partial_a\partial_b R_m)(p_+)V_\nu]\big).\end{aligned}$$

Apply (30) through order two and the same product inequality used above. It follows that

$$|\partial_{k_a}\partial_{k_b} F_{\mu\nu}(p,k)| \le C_\rho[x(p-k/2)^{-4} + x(p+k/2)^{-4}],$$

and (56) with $j = 4$ gives

$$\int_{[-\pi,\pi]^3} |\partial_{k_a}\partial_{k_b} F_{\mu\nu}(p,k)| \frac{\mathrm{d}p}{(2\pi)^3} \le C_\rho m^{-1}, \qquad k \in \mathbb{R}^3. \tag{60}$$

Taylor's formula with integral remainder, followed by (59), now makes the two momentum powers explicit:

$$\Pi^\eta_{\mu\nu}(K) = -\eta \int_{[-\pi,\pi]^3} \int_0^1 (1-s) \sum_{a,b} K_a K_b (\partial_{k_a}\partial_{k_b} F_{\mu\nu})(p, s\eta K)\,\mathrm{d}s\, \frac{\mathrm{d}p}{(2\pi)^3}. \tag{61}$$

This gives the bound $C_\rho(\eta/m)|K|^2$. For the gradient, differentiate (58), subtract its zero-momentum derivative using parity, and apply the first-order integral remainder:

$$\partial_{K_a}\Pi^\eta_{\mu\nu}(K) = -\eta \int_{[-\pi,\pi]^3} \int_0^1 \sum_b K_b (\partial_{k_a}\partial_{k_b} F_{\mu\nu})(p, s\eta K)\,\mathrm{d}s\, \frac{\mathrm{d}p}{(2\pi)^3}.$$

All differentiations are under an integral over a compact cube of a smooth function, since $m > 0$. Using (60) and $\eta/m = \bar{m}^{-1}$ proves (53). □

5.3. **Uniform bounds on the determinant coefficients.** We finish by pairing the polarization bounds with decay of the probe transform. This step needs no continuum evaluation of the two-photon kernel.

For each integer $j \geq 0$ there is $C_{\beta,j} < \infty$ such that

$$|h_{\eta,\nu}(K)| \leq C_{\beta,j}(1+|K|)^{-j}, \qquad |K|_\infty \leq \pi/\eta, \quad 0 < \eta \leq 1. \tag{62}$$

The constants are unchanged by any real translation of β. Here is the discrete integration by parts that makes this uniform in the mesh. Suppose a smooth function v is supported in a cube of side ℓ, and sum it over a translate of $\eta\mathbb{Z}^3$. The backward difference $\Delta_{\eta,\mu}v(y) = \eta^{-1}(v(y) - v(y - \eta e_\mu))$ satisfies $\|\Delta^j_{\eta,\mu}v\|_\infty \leq \|\partial^j_\mu v\|_\infty$, by its integral representation. Its support is enlarged by at most $j\eta$ in one direction, and the number of sampling points, multiplied by η^3, is at most $(\ell + j + 1)^3$. Summation by parts gives

$$[\eta^{-1}(1 - e^{-i\eta K_\mu})]^j \eta^3 \sum_y v(y)e^{-iK\cdot y} = \eta^3 \sum_y \Delta^j_{\eta,\mu} v(y) e^{-iK\cdot y}.$$

On $|\eta K_\mu| \leq 3\pi/2$ the modulus of the bracket is at least $0.3|K_\mu|$. Choose a coordinate attaining $|K|_\infty$. For $|K| \geq 1$ this is at least $(1 + |K|)/12$; for $|K| < 1$ use the point-count estimate directly. We thus obtain, on that momentum cube,

$$\left|\eta^3 \sum_y v(y)e^{-iK\cdot y}\right| \leq 12^j(\ell + j + 1)^3 \max(\|v\|_\infty, \max_\mu \|\partial^j_\mu v\|_\infty)(1 + |K|)^{-j}.$$

Apply this to $v = a^0_{\eta,\nu}$ on $\eta\mathbb{Z}^3 + \eta e_\nu/2$. Cube averaging has norm one on derivative sup norms, and a centered difference is an average of the corresponding first derivative. Consequently $\|\partial^j_\mu a^0_{\eta,\nu}\|_\infty \leq 2|\beta|_{C^{j+1}}$, while its support has side at most $D_\beta + 2\eta$. One may take

$$C_{\beta,j} = 2 \cdot 12^j (D_\beta + j + 3)^3 |\beta|_{C^{j+1}}.$$

This proves (62), with position-independent constants and no assumption on R.

Proposition 5.2. For each fixed ρ, $\bar{m}$ and β as above,

$$|c_2| \leq C_{\rho,\bar{m},\beta}, \qquad 0 < \eta \leq 1, \quad R \geq 1, \quad R/\eta \in \mathbb{N}, \quad R/\eta \geq 2.$$

If in addition $R \geq D_\beta + 1$, then $|c_4| \leq C_{\rho,\bar{m},\beta}$. The bounds are uniform in all real translations of β, and the coefficients have no dependence on e. ∎

Proof. Proposition 5.1 gives

$$|\Pi^{\eta,R}_{\mu\nu}(K)| \leq C_\rho(\bar{m}^{-3}R^{-4} + \bar{m}^{-1}|K|^2).$$

In (51) use the decay estimate with $j = 2$ for the quadrature term and $j = 3$ for the quadratic term. Since $R \geq 1$, both contributions are bounded by a constant times $(1 + |K|)^{-4}$; hence

$$|c_2| = |a_{\mathrm{ch}}| \leq 9C_\rho(\bar{m}^{-3}C^2_{\beta,2} + \bar{m}^{-1}C^2_{\beta,3})R^{-3} \sum_{K \in \mathcal{K}_{\eta,R}} (1 + |K|)^{-4}.$$

The last sum is bounded uniformly for $R \geq 1$. To see this, tile $\mathbb{R}^3$ with cubes of side $2\pi/R$ centered at $K \in (2\pi/R)\mathbb{Z}^3$. On each cube, $1 + |P| \leq (1 + \sqrt{3}\pi)(1 + |K|)$, so its volume times $(1 + |K|)^{-4}$ is bounded by a constant times the integral of $(1 + |P|)^{-4}$ there. Sum over the cubes; that integral over $\mathbb{R}^3$ is finite. Restricting the sum to $\mathcal{K}_{\eta,R}$ only decreases it. This proves the uniform second-order bound.

For the fourth coefficient, use (46) and the fourth-trace estimates (8), proved in section 4. The infinite-space trace $b(f)$ is finite by (41), and the two errors in (8) are bounded for $R \geq D_\beta + 1$ and $0 < \eta \leq 1$. Thus $|b_{\mathrm{ch}}|$ is bounded, and adding $3c_2^2$ gives the asserted bound for c_4. All constants in the proof depend on support sizes and derivative norms of β, not on its position. Independence of e follows directly from the external connection convention. □

Completion of the proof of theorem B. The fourth-trace assertions were proved in Section 4. The determinant symmetries and coefficient identities follow from Section 5, and Proposition 5.2 gives the second- and fourth-order bounds with the stated volume ranges. □

References

[26a] "Chamber fermion locality and local integration". Companion manuscript, 20 October 2026. 2026 (cit. on p. 8).

[26b] "Reflection positivity for chamber QED on the infinite lattice". Companion manuscript, 20 October 2026. 2026 (cit. on pp. 2, 13).

[26c] "The field-strength Osterwalder–Schrader axioms for QED in three dimensions". Comprehensive edition, 263 pp., 20 October 2026, before the modular integration revision. 2026 (cit. on pp. 1, 3).

[Gut15] L. Guth. *18.156 Lecture Notes: March 30, 2015*. Transcribed by Jane Wang; Theorems 4–7. Massachusetts Institute of Technology. 2015. URL: https://math.mit.edu/~lguth/Math156/notes/20.pdf (cit. on p. 10).

[HE36] W. Heisenberg and H. Euler. "Folgerungen aus der Diracschen Theorie des Positrons". *Zeitschrift für Physik* 98 (1936). English translation by W. Korolevski and H. Kleinert, 714–732. DOI: 10.1007/BF01343663. arXiv: physics/0605038 (cit. on p. 1).

[Pon25] R. Ponge. *Singular Values and Schatten Classes*. Introduction to Noncommutative Geometry, Chapter 4; slides 16 and 53–56. 2025. URL: https://raphaelponge.org/wp-content/uploads/2025/07/summer-chap4-1.pdf (cit. on p. 3).